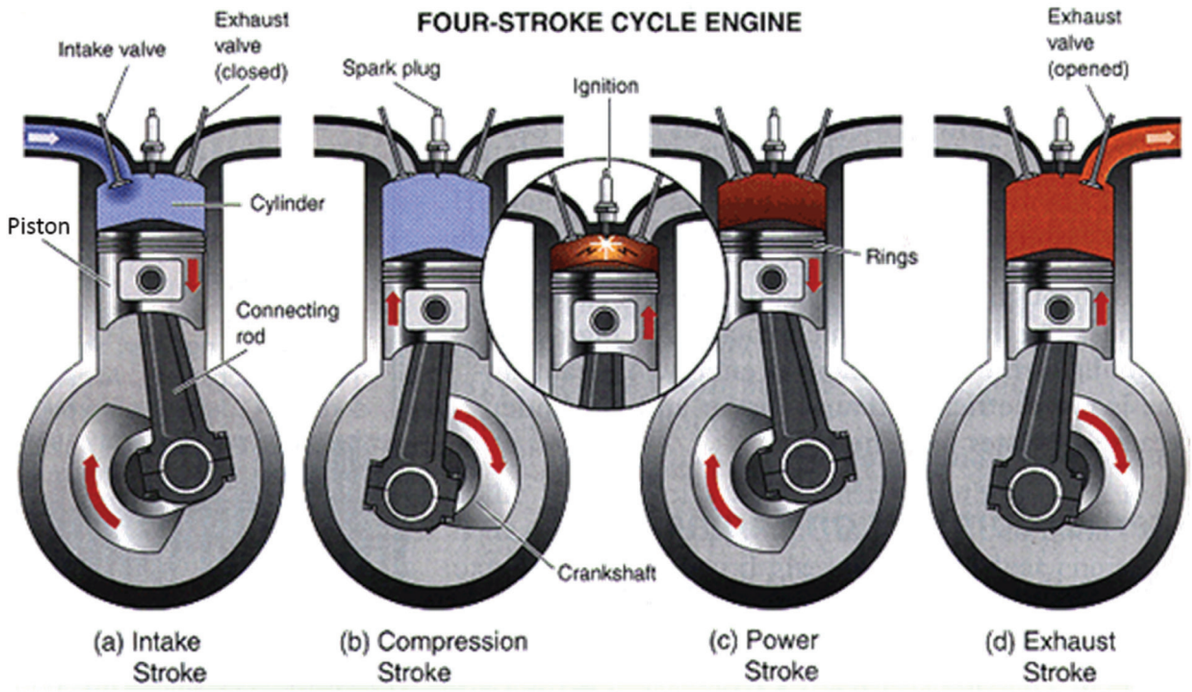


Figure 5.4 Working of Four Stroke Petrol Engine

a. Suction Stroke,

B. Compression Stroke,

C. Power Stroke And

D. Exhaust Stroke

The above four strokes will form a cycle. If the cycle of operations is completed in four strokes of the piston or two revolutions of the crankshaft or one revolution of the camshaft, then it is called as a four-stroke engine.

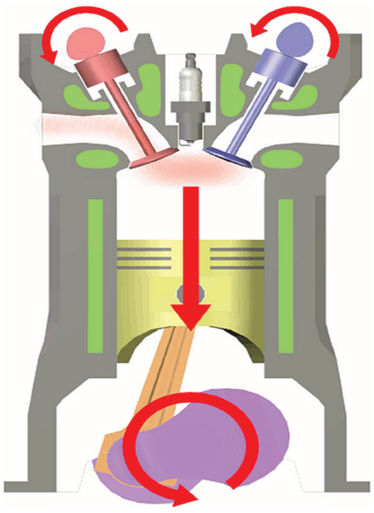


Figure 5.4(a)

a) Suction Stroke

During the suction stroke, the piston moves downward from Top Dead Centre to Bottom Dead Centre. The intake valve is open and exhaust valve is closed. This downward movement of the piston produces a partial void, or vacuum, in the cylinder, and air – fuel mixture rushes into the cylinder through the opened intake valve.

b) Compression Stroke

The charge taken into the cylinder during the suction stroke is compressed by the return stroke of the piston. The piston travels from bottom dead centre to top dead centre. During compression stroke both inlet and exhaust valves are in closed position.

AM_Unit_05.indd 98 12/20/2019 5:26:38 PM